Partitioning and scalability

Faculty of Informatics, Juraj Dobrila University of Pula

assoc. prof. Nikola Tankovic, PhD

Sample chapter for the proposed course plan.

1 Learning outcome

Identify how a partition key can create a hotspot.

Prerequisites: Lectures 04, 06, and 10.

2 Core idea

Partitioning assigns different data to different nodes. The chosen key affects both query routing and load distribution. Evenly dividing identifiers does not guarantee evenly dividing traffic.

3 Worked scenario

Registrations are partitioned by course identifier. One very popular course receives most requests during enrolment.

4 Discussion

Why might adding nodes leave the popular course as a bottleneck?

5 Recap

Evaluate the request distribution as well as the data distribution.